

A Monolithic Analog Radio-over-Fiber Uplink Transmitter with -100.8 to -98.3 dBm Effective Isotropic Sensitivity for mm-Wave Communications

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Abstract We present a monolithic analog radio-over-fiber uplink transmitter co-integrating LNAs, driver, and MZM on a CMOS SOI process. The transmitter achieves 26.1-dB gain, 8.2-dB noise figure, and -100.8 -dBm EIS, enabling DPD-free 1.1% and 2.25% EVM of 16-QAM and 32-QAM, respectively, meeting 3GPP requirements. ©2026 The Author(s)

Introduction

The deployment of next-generation cellular networks in the millimeter-wave (mm-wave) bands is primarily driven by the need for massive throughput. However, the limited propagation range of mm-wave signals necessitates a high density of base stations and small-cell architectures. To make such dense deployment economically and thermally viable, especially considering the integration of massive antenna arrays, lightweight and low-power remote units (RUs) are essential. Conventional digital radio-over-fiber (D-RoF) systems, while robust, require complex RUs with power-hungry high-speed analog-to-digital converters (ADCs) and digital signal processing (DSP), which severely limits RU scalability (Fig. 1(a)). While emerging hybrid schemes such as sigma-delta radio-over-fiber (S-DRoF) attempt to reduce downlink hardware complexity by moving quantization to the DU, they still force the RU to perform power-hungry high-speed continuous-time sigma-delta modulation (CT-SDM) for the uplink, which severely limits scalability in multi-antenna scenarios [1,2]. Conversely, analog radio-over-fiber (A-RoF) [3-7] remains the most promising candidate for massive front-end integration by centralizing all frequency conversion and DSP functions at the DU, thereby minimizing the RU power footprint and enabling direct digital beamforming (Fig. 1(b)).

However, the practical implementation of A-RoF uplinks is hindered by significant challenges in the microwave-photonic interface. A major bottleneck is the severe intrinsic link-budget penalty caused by the low efficiency of electrical-to-optical (E/O) conversion. This conversion loss, coupled with the relative intensity noise (RIN) of the laser and shot noise of the photodetector, places an extreme demand on the RF front-end for high gain and an ultra-low noise figure (NF). Furthermore, the inherent sinusoidal transfer function of the Mach-Zehnder modulator (MZM) imposes a fundamental trade-off between gain and linearity [8]. Prior arts [9-12] based on

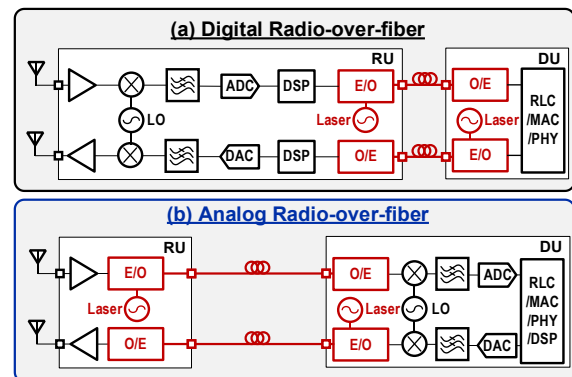


Fig. 1: (a) Digital radio-over-fiber. (b) Analog radio-over-fiber. RLC: radio link control; MAC: media access control; PHY: physical layer.

multi-chip solution exhibit low gain with NF >22 dB for mm-wave implementations [11,12], resulting in insufficient sensitivity that fails to meet the stringent 3GPP effective isotropic sensitivity (EIS) requirements. As a definitive 3GPP metric, EIS is critical for mm-wave uplink to evaluate the fundamental ability to detect and recover weak over-the-air signals.

In this paper, we address these limitations by presenting a monolithically integrated A-RoF uplink transmitter fabricated in a 45-nm RF-SOI process. By achieving a holistic co-design of low-noise amplifier (LNA), driver, and a MZM on a SOI substrate, we demonstrate a state-of-the-art performance that significantly improves the system-level noise and linearity balance, offering a scalable solution for future A-RoF cellular infrastructures.

Monolithic A-RoF Uplink Transmitter

Fig. 2 illustrates the schematics of the proposed monolithic A-RoF uplink transmitter. The RF signal from the antenna is processed through a 3-stage common-source LNA (M_1 to M_3) followed by a cascode driver (M_4), which then excites the 2-mm transmission-line (T-line) electrodes of the TW-MZM for optical intensity modulation. The continuous wave (CW) optical carrier is coupled via an on-chip grating coupler (GC). The modulated analog optical signal is then coupled into the

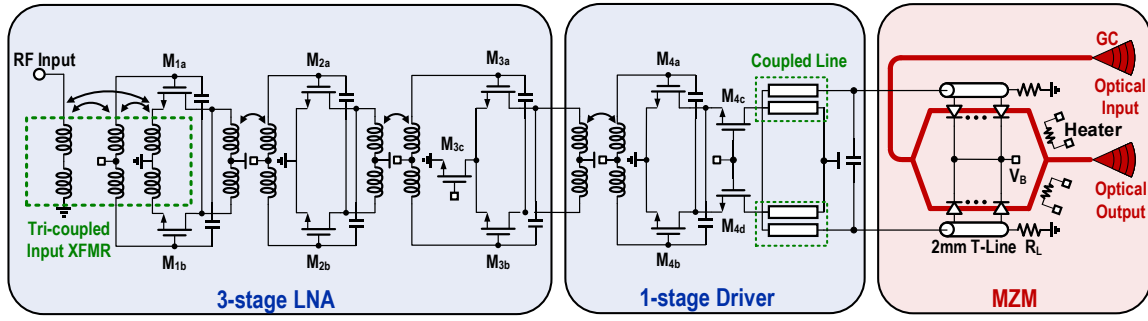


Fig. 2: The schematic of monolithic A-RoF uplink transmitter including 3-stage LNA, one-stage driver, and TW-MZM.

polarization-maintaining fiber via another GC.

The utilized 2-mm SiPh TW-MZM exhibit a half-wave voltage (V_{π}) of 8.25 V and a 36 GHz 3-dB bandwidth with T-section slow-wave structure. When terminated with a matching resistor (R_L), it presents a differential input impedance of 72 Ω . Dual-arm heaters are included for quadrature biasing.

To mitigate the DC supply wire-bonding parasitics, both the LNA and driver employ differential topologies with neutralization capacitors for enhanced stability and unilateral gain. The tri-coupled transformer (XFMR) based input matching network serves as an integrated balun while simultaneously implementing noise-canceling and Gm-boosting [13], resulting in a simulated minimum NF of 2.9 dB for the LNA-driver chain. For inter-stage matching, XFMR-based magnetically coupled resonators are utilized to extend the bandwidth, providing a total front-end gain of approximately 47 dB.

To seamlessly bridge this high-gain RF front-end with the optical modulator, a differential driving scheme is used and a coupled-line-based electrical-to-optical (E/O) matching network is implemented. This network performs a critical impedance transformation, mapping the MZM input impedance directly to the driver's optimal load derived from load-pull simulations to achieve precise load-line matching, thereby maximizing output linearity and power gain. Crucially, beyond fundamental impedance matching, the common-mode load impedance at second-harmonic frequency, established by the E/O matching network, is utilized to partially cancel the third-order intermodulation (IM3) components through the second-harmonic re-mixing, resulting in further enhancing the input 1-dB compression point (IP1dB) by 3 dB. Moreover, the coupled-line topology intrinsically provides essential DC isolation, completely decoupling the driver's power supply from the MZM's reverse-bias voltage.

Measurement Setups and Results

The prototype is fabricated in GlobalFoundries 45nm RF-SOI process. Fig. 3 and Fig. 4 show the measurement setups, chip micrograph, and

measurement results, respectively. A CW O-band signal from an Agilent 81600B laser is boosted by a Thorlabs praseodymium-doped fiber amplifier (PDFA) PDFA100X and fed into MZM via a 15-degree fiber array (FA) and on-chip GC, to compensate for the loss of the FA and GC. The modulated output is converted back to electrical signal using a photoreceiver (Thorlabs RXM40AF). The RF input signal is fed to the chip via a GSG probe. To establish the optimal operation, the MZM is biased at the quadrature point via a 0.55-V heater voltage with 16 mA of current, while its PN junctions are reverse-biased at 4 V.

Measured with the performance network analyzer (PNA) Agilent E8361A, the link achieves a peak gain of 26.1 dB at 25.4 GHz, outperforming recent state-of-the-arts. The 3-dB bandwidth spans 8.4 GHz (22.6–31 GHz), with $S_{11} < -10$ dB across 23.2–32.2 GHz. For linearity measurement, a 20-dB attenuator is inserted to lower the PNA output; the measured in-band peak IP1dB is -22 dBm at 23 GHz, corresponding to an OP1dB of 2.4 dBm and a calculated third-order input intercept point (IIP3) of -12.4 dBm. A gain-method based on the electrical spectrum analyzer (ESA)

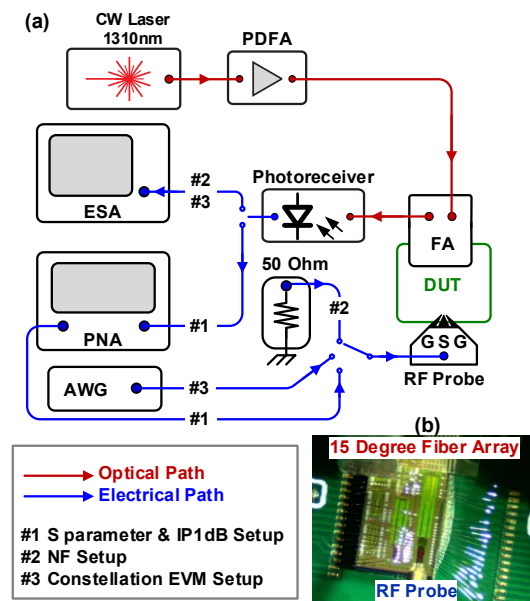


Fig. 3: (a) Setups for S-parameter (#1), IP1dB (#1), noise figure (#2), and EVM (#3) measurements. (b) Device under test (DUT) micrograph.

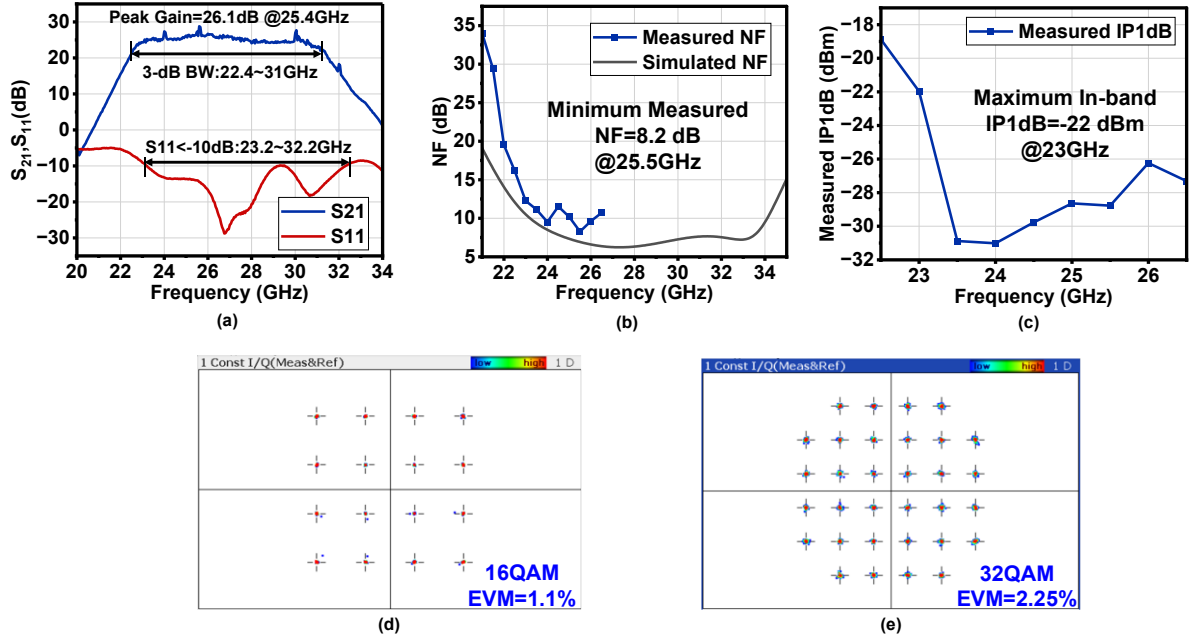


Fig. 4: Measured results of (a) s-parameter, (b) noise figure, (c) input 1-dB compression point, and the EVM of (d) 5MS/s 16QAM and (e) 10MS/s 32QAM signal at 25GHz carrier frequency.

R&S FSW26 is employed for NF measurement. The measurement result compared to prior arts shows a superior minimum NF of 8.2 dB, which is an end-to-end performance comprising noise contributions from electronics, laser, PDFA, MZM, and PD, and ensures sufficient EIS together with high directional gain of mm-wave antenna arrays for FR2 local area base stations for indoor and dense urban small-cell coverage [14]. To evaluate signal integrity, the arbitrary waveform generator (AWG) Keysight M8195A and the ESA R&S FSW26 are used. Without any DPD, the system achieves an RMS EVM of 1.1% for a 5MS/s 16-QAM signal and 2.25% for a 10MS/s 32-QAM signal at 25 GHz. Measurement bandwidth (of >10 MHz) and frequency (of >26 GHz) is currently limited due to the FSW26 capabilities. Table 1 compares our work with recent publications. Our

transmitter simultaneously optimizes the gain, linearity, and noise figure to meet the EVM and EIS requirements for 3GPP [14].

Conclusion

In conclusion, we demonstrated a monolithically integrated microwave-photonic transmitter that resolves the critical link-budget bottlenecks of A-RoF uplinks. By co-designing an LNA, driver, and MZM on a single SOI substrate, the architecture overcomes severe E/O/E conversion penalties. Validated at 25 GHz with DPD-free 16-QAM and 32-QAM operation, this low-noise, linear solution, to the best of the authors' knowledge, is the first chip-scale A-RoF uplink to meet 3GPP FR2 EIS requirements, paving the way for scalable, low-power remote units in next-generation cellular networks.

Tab. 1: Comparison with prior A-RoF uplink designs.

Performance	This work	[9]	[10]	[11]	[12]
Integration Type	Monolithic (LNA+Driver+MZM on SOI CMOS)	Hybrid (SiGe HBT LNA + SiPh MZM)	Hybrid (SiGe HBT LNA + SiPh MZM)	Hybrid (CMOS driver + LiNbO ₃ MZM)	Hybrid (CMOS driver + SiPh MZM)
Link Peak Gain(dB)	26.1	-1*	0*	3.3*	7*
Link 3-dB BW(GHz)	22.6~31	1~8*	1~6.4*	19~26.5*	15~25*
In-band NF(dB)	8.2~16	15.2~16.0*	13.4~13.9*	22*	35.1
Band n258 NF(dB)	8.2~10.7	N/A	N/A	N/A	N/A
Maximum IP1dB(dBm)	-22	N/A	N/A	N/A	N/A
Maximum IIP3(dBm)	-12.4*	5.5	22	4.8#	3
SFDR(dB*Hz ^{2/3})	94~100*	101~109	109~120	99#	85~92
DC Consumption(mW)	180	1650	1700	180	448
EIS(dBm)	-100.8~-98.3	-93.8~-84.8	-95.4~-86.9	-87	-73.9

*Estimated from figure, #the measured case, *calculated results, with IIP3=IP1dB+9.6dB. EIS is obtained with 12 dBi antenna and 0 dB SNR for 50MHz QPSK signal in 5G NR standard [14].

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